MIL-R-21098E
26 September 1985
SUPERSEDING
MIL-R-21098D
30 March 1983

MILITARY SPECIFICATION

REFRIGERATORS; FROZEN FOOD CABINETS; AND COMBINATION REFRIGERATOR-FROZEN FOOD CABINETS, MECHANICALLY REFRIGERATED, COMMERCIAL, SELF-CONTAINED, NAVAL SHIPBOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers commercial mechanical refrigerators, frozen food cabinets, and combination units including explosion-proof equipment for Naval shipboard use.

1.2 Classification. The refrigerating equipment shall be of the following types, styles, sizes, and classes as specified (see 6.2).

Type I - Refrigerator
Type II - Frozen food cabinet
Type III - Combination refrigerator-frozen food cabinet

Style 1 - Vertical cabinet
Style 2 - Horizontal cabinet

Sizes - Various sizes for each type and style (see tables I, II, III, IV, V, and VI)

Class A - Explosion proof
Class B - Non-explosion proof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4110

DISTRIBUTION STATEMENT A. Approved for public release; distribution unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-F-1421	-	Flourocarbon Refrigerants
QQ-A-250	-	Aluminum and Aluminum Alloy Plate and Sheet General Specification For
QQ-B-650	-	Brazing Alloys, Copper, Copper-Zinc, and Copper-Phosphorous
QQ-B-654	-	Brazing Alloy, Silver
QQ-S-766	-	Steel Plates, Sheets, and Strip, Corrosion-Resisting
QQ-W-423	-	Wire, Steel, Corrosion-Resisting

MILITARY

MIL-C-2212	-	Controller, Electric Motor A.C. or D.C., and Associated Switching Devices
MIL-P-2693	-	Packing Material, Cold Storage Door Gasketing, Non-watertight
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-12204	-	Solder, Lead-Tin Alloy
MIL-T-12323	-	Refrigerators and Related Equipment, Packaging and Packing of
MIL-P-15024/6-		Plates, Identification, Equipment
MIL-M-17556	-	Motor, Direct-Current, Fractional HP (Shipboard Use)

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1881	-	Brazing, Silver; General Process For

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 62 - Casting, Composition Bronze or Ounce Metal
B 280 - Seamless Copper Tube for Air Conditioning and Refrigeration Field Service
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1-18.076 to 18.092 - Motors for Hermetic Refrigeration Compressors

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

UNDERWRITERS LABORATORIES, INC.

UL 303 - Refrigeration and Air Conditioning Condensing and Compressor Units
UL 471 - Commercial Refrigerator and Freezers
UL 873 - Temperature-Indicating and Regulating Equipment
UL 886 - Outlet Boxes and Fittings for Use in Hazardous Locations, Class I, Groups A, B, C, and D, and Class II, Groups E, F, and G

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingston Road, Northbrook, IL 60062.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code, Article 500, Hazardous Locations

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02169.)

NATIONAL SANITATION FOUNDATION (NSF)

No. 7 - Food Service Refrigeration and Storage Freezer

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Road, P.O. Box 1468, Ann Arbor, MI 48106.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials, parts, and components shall be in conformance with the requirements specified herein. All materials, parts, and components forming a part of the finished product shall be new and of adequate size or capacity for the intended purpose. All materials shall be free from any defects that might affect the serviceability or appearance of the finished product (see 6.3).

3.1.1 Hazardous materials. Materials which are capable of producing dangerous gases or other harmful effects under conditions (including fire) encountered in Naval shipboard service shall not be used without specific approval of the contracting officer. Magnesium and its alloys shall not be used. No instruments containing mercury shall be used in the manufacture or testing of any equipment destined for installation on a ship.

3.2 Corrosion prevention.

3.2.1 Protection against electrolytic or galvanic corrosion. Equipment shall be constructed in accordance with MIL-STD-889 to provide protection against electrolytic or galvanic corrosion. Joints between dissimilar metals which may become wetted in service shall be protected against electrolytic or galvanic corrosion by proper selections of materials, plating, isolation, insulation, paint, area relationship, gaskets, or other means.

3.2.2 Corrosion-resisting materials. Corrosion-resisting steel, copper alloys, and aluminum alloys are satisfactory corrosion-resisting materials. Where aluminum is specified for the cabinet, it shall conform to QQ-A-250. Where corrosion-resisting steel is specified, except for shelves (see 3.5.4.4), it shall conform to class 302 or 304 of QQ-S-766. Corrosion-resisting steel, when fabricated by any method that tends to reduce corrosion-resisting properties, shall be normalized to restore those properties before being assembled in any unit.

3.2.3 Surface protection. Parts other than corrosion-resisting materials shall be protected against corrosion with chemicals, electrolytic processes, plating, or suitable paints and enamels. Cadmium, antimony, bismuth, lead, or other toxic materials shall not be used in any way where it may come into contact with food, ice, or other materials for human consumption. The following methods, while not restrictive, are acceptable corrosion-protection methods when properly applied:

a. Hot-dipped galvanized.
b. Hot phosphate process with rust-inhibiting paint.
c. Electroplated cadmium followed by dichromate, or hot-phosphate treatment or water-dip lacquer.
d. Caustic dip, "anodized" coated with phenolic resin baked on, or covered with lacquer for aluminum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

3.2.4 Fasteners. All screws exposed in final assembly shall be corrosion-resisting steel. All others shall be of the same material being joined or supported. Self-tapping machine screws or standard machine screws with locknuts or tapped holes are acceptable. Sheet metal screws are not acceptable.

3.2.5 Finishes. Corrosion-resisting steel, where exposed in final assembly, except inside the machinery compartment, shall have a number 3 finish as specified in QQ-S-766, or better. When aluminum alloy is specified, it shall be clean, bright, smooth, flat, and free from defects and contaminants. Aluminum finishes for the inner and outer surfaces of the cabinet shall be anodized in accordance with type I, class 1 of MIL-A-8625. No discoloration such as a yellowish tone will be accepted for a finished aluminum surface.

3.2.6 Thermal insulation. The thermal insulation shall be a nonmoisture absorbing material and homogeneous throughout. The insulation "K" factors shall be as specified below using a 75° Fahrenheit (F) mean temperature and expressed in terms of British thermal units (BTU) per hour per square foot per square inch of thickness per degree F temperature difference. Loose fill type insulation will not be acceptable. The insulation shall be installed with sufficient thickness so that condensation will not occur on any external surface other than door mullions on units having two or more doors, when operating under conditions of 100°F ambient temperature with relative humidity of not less than 50 percent (see 4.4.1). Electrical heating may be used around the door to eliminate external moisture. The insulation shall be of a type that is not combustible or will not support combustion and will not release toxic fumes when exposed to high temperature.

3.2.6.1 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial rated "K" factor of not more than 0.15, and a core density of not less than 1.7 pounds per cubic foot when tested as specified in 4.4.9.

3.2.6.2 Pre-expanded insulation. Pre-expanded insulation shall be polystyrene or polyurethane material having a manufacturer's commercial rated "K" factor of not more than 0.26 for polystyrene and and 0.15 for polyurethane with a core density of not less than 1.7 pounds per cubic foot when tested as specified in 4.4.9.

3.2.6.3 Semirigid insulation. Semirigid insulation shall be "B" fiber or finer industrial type fiberglass with manufacturer's commercial rated density of at least 1.0 but not more than 3.0 pounds per cubic foot and shall be held together with a thermosetting binder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

3.3 Interchangeability. Similar parts or corresponding apparatus furnished on the same contract or order or built to the same drawings, shall be strictly interchangeable without the necessity of further machining or hand fitting of any kind.

3.4 Design. Each refrigerator, frozen food cabinet, and combination unit shall consist of an insulated cabinet, refrigerating equipment, controls and accessories all enclosed within a metal cabinet. Unless otherwise specified herein the design and construction of the equipment shall conform to the requirements of UL 471 and NSF No. 7. The condensing unit shall be located below the refrigerated compartment in the style 1 vertical cabinet units. Unless otherwise specified (see 6.2), the condensing unit shall be located on the left side of the style 2 horizontal cabinet units. A separate condensing unit shall be used for each low temperature and high temperature compartment in the combination unit. The units shall be factory assembled, complete, and self-contained with the refrigeration equipment dehydrated and charged with the necessary operating quantities of refrigerant and oil. Each unit shall be ready for operation after removal of shipping protection, opening of valves, and connection to electrical power. Each refrigerator compartment shall be equipped with interior lighting automatically controlled by a switch or switches activated by the opening and closing of any door of the compartment. Interior lighting shall not be provided for low temperature (freeze) compartments or for refrigerator compartments designed for explosion-proof applications.

3.4.1 Accessibility. Valves, controls, and equipment subject to service and repair shall be readily accessible for servicing from the front of the unit and, when located in the machinery section, shall be reached through a removable front panel. Where panels are provided for access to equipment, machine screws or fastener type hardware may be used for retention.

3.4.2 Ventilation. Adequate ventilation shall be provided for condensing unit equipment. Where ventilation is provided in the front, the opening shall be die formed, either louvered or slotted type.

3.4.3 Inclination. The equipment shall operate satisfactorily when inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other (see 4.4.2).

3.4.4 Explosion-proof class A. Class A refrigerators, frozen food cabinets or combination refrigerator-frozen food cabinets shall be explosion-proof, conforming to the requirements of NFPA No. 70, article 500 for class 1, division 1 locations, group C atmospheres (see 4.4.3).

3.4.5 Compliance. The contractor shall submit to the contracting officer or his authorized representative, satisfactory evidence that the machines he proposes to supply under this document, meet the applicable requirements of UL No. 471 and NSF No. 7 as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

3.4.5.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL listing mark or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that machines offered have been tested and conform to UL 471.

3.4.5.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

a. A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished machine, or;

b. A certification for the machine issued by NSF under their special one-time contract evaluation/certification service, or;

c. A test report, acceptable to the contracting officer with the advice of the Army Surgeon General, from an independent testing laboratory indicating that the items have been tested and conform to the specified NSF Standard No. 7.

3.5 Construction.

3.5.1 Frame. The cabinet and condensing unit sections shall have an integral structural frame. The frame shall be constructed of welded members and treated for corrosion-resistance after fabrication. The frame shall be a full perimeter frame to provide structural rigidity to the cabinet and provide support for major parts and components. When polyurethane foamed-in-place insulation is used, the integral structural frame may be eliminated from the front of the cabinet, however, the bottom integral structural frame shall be provided for support of the legs. All types of cabinets shall be provided with 7-1/2 inch long mounting legs. This leg height shall not be included in the height requirements as specified in tables I thru VI. The removable mounting legs shall be secured to the bottom four corners of the cabinet, not more than 2 inches in from the corner exterior surfaces (i.e. front and sides and back and sides) with hardware deemed necessary and as generally provided by the manufacturer. For cabinets where the corner legs are spaced more than 48 inches apart, a pair of center legs shall be provided. The bottom of each mounting leg shall have a 11/16 inch hole to accept a 5/8 inch bolt for securing and shipping purpose.

3.5.2 Condensing unit. The condensing units shall be designed for operation with refrigerant 12 (dichlorodifluoromethane or refrigerant 22 (monochlorodifluoromethane) conforming to BB-F-1421. Each unit shall consist of a compressor, condenser, receiver (if required), fan, motor, and accessories mounted on a fabricated steel frame or base. Each condensing unit shall conform to the requirements of UL 303. Each refrigerated compartment shall be served by its own condensing unit. The capacity of each condensing unit shall be adequate for the intended application. Motor pulleys and compressor flywheels on open compressor condensing units for explosion-proof applications shall be of nonsparking materials. Drive belts shall contain sufficient electro-conductive material to prevent the development of electrostatic charges.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

3.5.2.1 Compressor. The compressor shall be of the open type for d.c. or of the hermetic type for a.c. Provision shall be made for adequate lubrication of all rubbing and wearing surfaces, including operation under shipboard conditions of inclination and rolling.

3.5.2.1.1 Open type. The open compressor shall be of the vertical, single acting, reciprocating type. The shaft seal shall be replaceable in its entirety without the necessity of removing or replacing the crankshaft. The suction and discharge shut-off valve connections to the compressor shall be of the flange type.

3.5.2.1.2 Hermetic type. The hermetic type compressor shall consist of a compressor and a motor enclosed in a gas-tight shell. The hermetic compressor for explosion-proof applications shall have adequate seals around electrical terminals, be provided with a suitable wiring enclosure, and be of a type or design that is Underwriters Laboratories approved as explosion-proof for National Electrical Code, group C and D atmosphere.

3.5.2.2 Condenser. The condenser shall be forced air cooled and constructed of finned tubes. Fins shall be soldered, welded, or mechanically bonded to the tubes and shall be not less than 0.010 inch thick. The condenser shall be treated for corrosion-resistance after fabrication. The fan shall be located as near as is practical to the center of the condenser face and, where necessary for uniform air flow, a metal or a plastic fan shroud shall be provided. Condensers for explosion-proof applications shall be designed for natural convection and shall be adequate for the intended application in ambient temperature up to 100°F.

3.5.2.3 Liquid receiver. A receiver shall be furnished when an open type compressor is used and may be used in a hermetic system. The liquid receiver, when furnished, shall be such that it has an internal volume at least 25 percent greater than the liquid volume of the complete refrigerant charge. The receiver shall contain a fusible plug for protection against excessive temperatures. The plug shall melt at a temperature not lower than 200°F and not higher than 300°F. Where receivers are not furnished, the pressure relief device shall be incorporated in the condenser.

3.5.2.4 Refrigerant piping. Piping shall conform to UNS No. C12200 as specified in ASTM B 280. All fittings shall be of wrought copper or forged brass. All fittings and joints, where practical, shall be brazed according to MIL-STD-1881 with silver brazing alloy as specified in QQ-B-654 or copper-phosphorous brazing alloy as specified in QQ-B-650. Connections shall be kept at a minimum and, where tubing joints are necessary, couplings shall be used. Piping adjacent to frame members or to other components or parts shall be rigidly supported to prevent movement and rubbing as a result of environmental vibration. Suction lines to a compressor within machinery compartment shall be insulated to prevent condensation. Piping systems shall withstand the pressure and leak test specified in 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

3.5.2.4.1 System integrity. The entire refrigerant circuit shall be secure and leak free when tested as specified in 4.4.5.

3.5.2.5 Dehydrator. A dehydrator shall be provided in the refrigerant circuit. Each dehydrator shall be equipped with an auxiliary screen or other protective means at the dehydrator outlet to prevent passage of dehydrating agent in the event of rupture of the cartridge screen outlet. The desiccant shall be silica gel or activated alumina or a type satisfactory to the command or agency concerned.

3.5.2.6 Heat exchanger. A heat-interchanger shall be provided in the suction line. Bonding of the liquid refrigerant line to the suction line to provide adequate liquid subcooling is considered satisfactory as a heat-interchanger.

3.5.2.7 Expansion devices. A capillary tube shall not be used as the refrigerant expansion device when an open type compressor is employed. When a refrigerant expansion valve is used, the valve shall be a pressure limiting type. Where an expansion valve is used and the compressor motor is not a type that will start the compressor against the system pressure differential, the expansion valve shall be provided with a bleed port to equalize the suction and discharge pressures during shut down.

3.5.3 Electrical equipment. Unless otherwise specified, the electrical equipment shall conform to NEMA MG1-18.076 to 18.092 and UL 873, as applicable. No portion of the electrical circuit shall be grounded. The frames or enclosures of all electrical components shall be grounded to the cabinet to eliminate hazard from shorts or grounds within the equipment. Equipment and terminals shall be adequately protected against moisture, dirt, mechanical damage and accidental contact by personnel. Equipment shall be furnished only for the following power supply as specified (see 6.2):

115 volts single phase, 60 cycle a.c. with hermetically sealed compressor.
115 volts d.c. with open type compressors.

3.5.3.1 Compressor motors. Direct current motors shall comply with MIL-M-17556. Compressor motors shall be not less than 1/4 horsepower. The motor shall be constant speed, continuous duty and shall be protected against overload by a thermal protector built into the motor or mounted on the motor compressor unit. The temperature rise of the hermetically sealed motor shall not exceed 70°C. In addition to the above, d.c. compressor motors shall conform to the following:

Horsepower - 1/4 (minimum)
Voltage - 115 Volts single phase, 60 cycle a.c. or 115 Volts d.c.
Service Factor - 1.25 to 1.35.
Classification - Class 1, Group C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

Enclosure - Dripproof or explosion proof, as specified (6.2).
Bearings - Ball.
RPM - 1725 nominal full load.
Winding - Shunt (or stabilized shunt), constant speed.
Ambient temperature - 50°C.
Insulation - Class A.

Under maximum operating conditions the motor temperature rise shall be not more than 15°C above corresponding temperature rise (see 4.4.8) permitted in the motor specification.

3.5.3.1.1 Motor controller direct current unit. The d.c. unit shall be provided with a manual controller conforming to MIL-C-2212. The controller shall provide overload protection. The controller enclosure shall be dripproof or explosion-proof as specified (see 6.2).

3.5.3.2 Fan motor. When a separate motor is used to drive the condenser cooling fan or the evaporator fan, the fan motor enclosure shall be dripproof. The motor shall be of a type that has a sealed or enclosed lubricant system and does not require bearing lubricant to be changed or supplemented periodically.

3.5.3.3 Wiring. The wiring between the electric devices (motors, motor controllers, control devices) shall conform to the requirements of UL 471 or UL 303, as applicable. The wiring shall be complete and the equipment shall be ready for connection of the incoming power cable at installation. A terminal block for three-conductor power cable shall be provided for the incoming power cable.

3.5.3.3.1 Lighting circuit. The lighting circuit shall consist of a plunger type switch, receptacle and incandescent lamp, and the necessary wiring. A corrosion-resistant type guard shall be provided to protect the lamp assembly against breakage.

3.5.3.3.2 Explosion-proof electricals. When explosion-proofing is required, all switches, relays, terminal blocks, wiring connections or junctions shall be installed in explosion-proof enclosures conforming to UL 886. Thread engagement of conduit, connectors, and covers shall be in accordance with UL 886. No holes shall be drilled through the enclosure for mounting components in the enclosure or for fastening the enclosure to the equipment. Holes for mounting components within the enclosure shall be bottomed leaving a minimum of 1/8 inch wall thickness. Enclosure covers shall be provided with a lock wire or similar sealing device to prevent inadvertent opening of the explosion-proof enclosure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

3.5.4 Cabinets. Refrigerators, frozen food cabinets, and combination refrigerator-frozen food cabinets shall have exterior and interior liners of corrosion-resistant steel unless otherwise specified (see 6.2). Corrosion-resistant steel for cabinet exteriors shall have a nominal thickness of 0.035 inch. Corrosion-resistant steel for cabinet liners shall have a nominal thickness of 0.031 inch. Where aluminum cabinets are specified, the nominal thickness of the aluminum for both interiors and exteriors shall be 0.040 inch. Except for breaker strips, all inner and outer surfaces of the refrigerated compartment(s) shall be made of corrosion-resistant steel or aluminum alloy when specified (see 6.2). Edges of sheets shall not be visible on the front or sides of the cabinet exterior, except for the case back, which shall have all seams and openings to case sides sealed airtight to form permanent vapor seal for the insulation. Cabinet interiors (liners) shall not have any cracks and shall be of constructed with either continuous welded or spot welded seams. Continuous welded seams shall present a smooth appearance with all discoloration removed. Spot welded seams shall have sufficient welds to assure structural rigidity equal to that of a continuously welded seam. Spot welded seams shall be continuously soldered with soft solder or filled with a silicone sealant, provided the materials and methods meet NSF requirements. Soft soldered seams shall be corrosion-resistant and shall, consistent with good industrial practice in the manufacture of same, be free of cadmium, antimony, bismuth, or other toxic materials. Silicone sealant, if applied shall be continuous and free from imperfections and runs. All seams shall be smooth and free from cracks and imperfections in bonding to the liner material. Aluminum shall be soldered with type I or II, composition B, conforming to MIL-S-12204. Flux shall not be retained by the seam after soldering is completed. Residual flux shall be inhibited by a hot water rinse. Solder shall not be used on surfaces of equipment with which unpacked, edible food may come in contact during the normal use of equipment. Non-metallic thermal insulating breaker strips shall be provided around the cabinet door opening. Hinge and latch tapping plates shall be provided and shall be an integral part of or welded to the cabinet frame. Tapping plates shall have threaded holes or be provided with retained nuts. The holes and threading shall be as required for the hinge and latch fastening screws. The thread engagement shall be not less than the diameter of the hinge and latch fastening screw plus one thread. The bottom of each refrigerated compartment shall be adequate to support a minimum weight of 0.50 pounds per square inch (psi). The complete exterior and interior of the refrigerator shall be free from scratches, dents, blemishes, or other defects. The shell and liner shall be reinforced to minimize flexing.

3.5.4.1 Thermometer. A dial or digital thermometer(s) shall be provided on the front of the cabinet for reading the temperature of each refrigerated compartment. The thermometer(s) shall not be mounted on the door(s). The thermometer dial size shall not be less than 2 inches with an operating range of minus 40°F to plus 60°F, with figure intervals of 10°F and 2°F minor graduations. The accuracy of the thermometer shall be within plus or minus one percent of the thermometer range. Thermometers containing mercury are not acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

3.5.4.2 Doors. Door exteriors shall be of the same material as the cabinet exterior. Metallic door liners shall be of the same material as the refrigerated compartment liner. The doors shall have an internal welded frame with hinge and latch tapping plates integral with or welded to the frame. When polyurethane foamed-in-place insulation is used the internal welded frame may be eliminated. The hinge and latch tapping plates shall be welded to the interior door frame. The doors shall have replaceable, cushion-type gaskets conforming to the requirements of type A gasket material of MIL-P-2693. The gaskets shall be close fitting and installed to provide a tight seal between the door and cabinet (see 4.4.6).

3.5.4.3 Hinges and latches. The door hinges and latches shall be heavy duty commercial type. Hinges and latches shall be chrome plated. Steel may be used for the tongue, striker insert, pins, and similar items. High impact plastic may be used for the striker insert. Cast brass, where used, shall conform to ASTM B 62. Latches shall be adjustable, arranged for locking, and provided with two keys where the lock is integral. For explosion-proof applications, the latch and striker shall be of nonsparking materials. Unless otherwise specified (see 6.2), the location of hinges when facing the cabinet shall be as follows:

	Mounting of hinges
a. For single door	Right hand
b. For multiple doors, one above the other	Right hand
c. For multiple doors, side by side	On opposite sides

3.5.4.4 Shelves. Five shelves for upright and two per door for horizontals shall be provided and shall be made of corrosion-resistant steel wire conforming to form I, composition 302 or 304, condition A of QQ-W-423. Wire used in construction shall be not less than the following diameters: (a) Frame wires - 0.375 inch; (b) cross wires - 0.125 inch (3/4 inch centers); (c) centers reinforcing wires - 0.380 inch. A front guard wire of 0.257 inch shall provide a lip approximately 3/4 inch above the shelf surface. Joints shall be welded, cleaned and sharp corners removed. Alternate shelves of stainless steel sheet, slotted or perforated, conforming to class 302 or 304, condition A of QQ-S-766 and equivalent configuration may be substituted for the wire type. The provided shelves shall be spaced on approximately 10-inch centers. Provision shall be made to prevent accidentally pulling out of the shelf. Shelf mounting brackets shall be made of corrosion-resistant steel or non-ferrous alloy, chrome plated, and adjustable on 2-inch centers. Shelves, supports, and mounting brackets shall be designed to support a minimum weight of 0.33 psi.

3.5.5 Type I, refrigerator. The refrigerator shall consist of an insulated compartment with cooling coil, condensing unit, controls, and accessories, all enclosed within a metal cabinet.

3.5.5.1 Capacity. The capacity and dimensional limitations of the refrigerator shall conform to table I for style 1 refrigerator and table II for style 2 refrigerators.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

TABLE I. Capacity of style 1 refrigerators

Size (number)	Net capacity food storage (cubic-feet) (minimum)	Maximum overall dimensions Width (inches)	Depth (inches)	Height 2/ (inches)	Number of doors
5	4	33	27 1/	64	1
10	9	32	29 1/	66	1
15	14	39	29 1/	72	1 or 2
20	19	51	29 1/	74	2
30	29	58	32	74	2
40	39	68	34	74	2 or more
50	49	84	34	74	2 or more
65	64	86	34	74	2 or more
85	84	112	34	74	2 or more

1/ Depth shown is maximum overall with door installed. With door and attaching hardware removed, maximum net depth shall not exceed 25-3/4 inches so that the refrigerator may be passed through a 26 inch by 66 inch opening.

2/ Does not include the leg height.

TABLE II. Capacity of style 2 refrigerators

Size (number)	Net capacity food storage (cubic-feet) (minimum)	Maximum over-all dimensions Width (inches)	Depth 1/ (inches)	Height 2/ (inches)	Number of doors
5	4.5	40	28	28	1
10	8	49	28	28	1
20	18	84	28	28	2
30	27	115	28	28	2 or more

1/ Depth shown is maximum overall with door installed. With door and attaching hardware removed, maximum net depth shall not exceed 25-3/4 inches so that the refrigerator may be passed through a 26 inch by 66 inch opening.

2/ Does not include the leg height.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

3.5.5.2 Drip pans. Evaporator drip pans shall be provided and shall be made of corrosion-resisting steel nominal 0.038 inch thick or aluminum alloy nominal 0.040 inch thick. Seams shall be watertight after welding, spot welding or silver soldering. Spot welded seams shall be sealed with silicone sealant (see 3.5.4). Drip pans shall have a drain fitting in the bottom of the pan and a drain line shall be provided. The drain line shall lead outside the cabinet and terminate in a fitting suitable for continuing the line to a vented deck drain.

3.5.5.3 Cooling unit. The cooling unit shall be of the forced-circulation type. Evaporators for explosion-proof refrigerators shall be natural convection type.

3.5.5.3.1 Forced-circulation type. The forced-circulation type cooling unit shall consist of cooling coil, fan with motor, and inclosing casing. The cooling coil shall be of finned tube construction using copper tubes. The coil tube sheets shall be aluminum, copper, or brass. The coil shall be electro-tin plated or plasticized after fabrication. The fan shall be of the propeller type. The fan and hub shall be made of brass (electro-tin plated), aluminum, corrosion-resistant steel, or plastic. The fan and motor shall be direct-mounted. The grill opening, shroud, and inclosing casing shall be made of corrosion-resistant steel or aluminum alloy.

3.5.6 Type II, frozen food cabinets. The frozen food cabinet shall consist of an insulated compartment, a condensing unit, a cooling unit, controls and accessories inclosed within a metal cabinet.

3.5.6.1 Capacity. The cabinet shall have capacities and dimensional limitations in accordance with table III for style 1 cabinets and table IV for style 2 cabinets.

TABLE III. Size and capacity of style 1 frozen food cabinets

Size (number)	Net capacity food storage (cubic-feet) (minimum)	Maximum over-all dimensions Width (inches)	Depth (inches)	Height 2/ (inches)	Number of doors
5	4	33	27 1/	64	1
10	9	32	29 1/	66	1
15	14	39	29 1/	72	1 or 2
20	19	51	29 1/	74	2
30	29	58	32	74	2
40	39	68	34	74	2 or more

1/ Depth shown is maximum overall with door installed. With door and attaching hardware removed, maximum net depth shall not exceed 25-3/4 inches so that the cabinet may be passed through a 26 inch by 66 inch opening.

2/ Does not include the leg height.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

TABLE IV. Size and capacity of style 2 frozen food cabinets

Size (number)	Net capacity food storage (cubic-feet) (minimum)	Maximum over-all dimensions Width (inches)	Depth 1/ (inches)	Height 2/ (inches)	Number of doors
5	4.5	40	28	28	1
10	8	49	28	28	1
20	18	84	28	28	2
30	27	115	28	28	2 or more

1/ Depth shown is maximum overall with door installed. With door and attaching hardware removed, maximum net depth shall not exceed 25-3/4 inches so that the cabinet may be passed through a 26 inch by 66 inch opening.

2/ Does not include the leg height.

3.5.6.2 Cooling unit. The cooling unit shall be a refrigerated inner liner with the refrigerating coils incorporated in or attached to the walls of the cabinet. Blower type cooling units and refrigerated plate type shelves are not permitted.

3.5.7 Type III, combination refrigerator - frozen food cabinet. The unit shall consist of a metal cabinet with two insulated compartments separated by an insulated wall. Each compartment shall have its own cooling unit, controls, and accessories.

3.5.7.1 Capacities. The unit shall have capacities and dimensional limitations in accordance with tables V and VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

TABLE V. Size and capacity of style 1 combination refrigerator-frozen food cabinets

Net capacity food storage			Maximum over-all dimensions			
Size (number)	Total (cubic-feet) (minimum)	Low temp. compartment (cubic-feet) (minimum)	Width (inches)	Depth (inches)	Height 2/ (inches)	Number of doors
10	9	3	31	29 1/	66	1 or 2
15	14	6	39	29 1/	72	2
20	19	8	51	29 1/	74	2
30	29	13	64	32	74	2
40	39	18	68	34	74	2 or more

1/ Depth shown is maximum overall with door installed. With door and attaching hardware removed, maximum net depth shall not exceed 25-3/4 inches so that the refrigerator-frozen food cabinet may be passed through a 26 inch by 66 inch opening.

2/ Does not include the leg height.

TABLE VI. Size and capacity of style 2 combination refrigerator-frozen food cabinets

Net capacity food storage			Maximum over-all dimensions			
Size (number)	Total (cubic-feet) (minimum)	Low temp. compartment (cubic-feet) (minimum)	Width (inches)	Depth 1/ (inches)	Height 2/ (inches)	Number of doors
20	18	8	84	28	28	1
30	27	13	115	28	28	2 or more

1/ Depth shown is maximum overall with door installed. With door and attaching hardware removed, maximum net depth shall not exceed 25-3/4 inches so that the refrigerator-frozen food cabinet may be passed through a 26 inch by 66 inch opening.

2/ Does not include the leg height.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

3.5.7.2 Cooling units. The cooling unit in the high temperature compartment shall be the forced circulation type. The cooling unit in the low temperature compartment shall be a refrigerated inner-liner with the refrigerating coils incorporated in or attached to the walls of the compartment. Blower type cooling units and refrigerated plate type shelves are not permitted.

3.5.8 Operating requirements (types I, II and III).

3.5.8.1 High temperature compartment. The mechanical cooling apparatus shall automatically produce and maintain an average refrigerated compartment temperature of 37° ± 3°F when the ambient temperature is 100°F, without food or ice load. Under these conditions, the condensing unit shall operate not more than 90 percent of the time (see 4.4.7).

3.5.8.2 Low temperature compartment. The mechanical cooling apparatus shall automatically produce and maintain an average compartment temperature of minus 5° ± 3°F without food load when the ambient temperature is 100°F. Under these conditions, the condensing unit shall operate not more than 90 percent of the time (see 4.4.7).

3.5.8.3 Overload condition. The unit shall operate satisfactorily with all doors wide open (see 4.4.8) in an ambient temperature of 120°F without tripping the compressor motor overload device or without breakdown. The motor temperature rise shall be not greater than that specified in 3.5.3.1.

3.5.8.4 Automatic operation. The automatic operation of each compressor shall be controlled by means of a temperature control switch. The switch shall have a knob adjustment to change cabinet temperatures. The controls shall be set to maintain a temperature of 37° ± 3°F in the high temperature compartment and minus 5° ± 3°F in the low temperature compartment. The temperature controls shall have a minimum range of 35° to 50°F for the high temperature compartment and a minimum range of plus 5° to minus 10°F for the low temperature compartment.

3.5.8.5 Alarm switches. When specified (see 6.2), additional thermostatic switches shall be provided to activate an external alarm system. The switches shall be mounted in the machinery compartment of the refrigerator. The thermal bulbs of the switches shall be located in the refrigerator or freezer compartment to sense the compartment temperature. Physical protection shall be provided for the thermal bulbs if they are located where they could be damaged by supplies stored in the refrigerator or freezer. The switches shall be identified as high temperature alarm switch and low temperature alarm switch. The identification shall be by permanently attached label, decal, or stencil. The temperature alarm switches shall be normally closed and shall have a range of differential suitable for the following settings:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

a. Refrigerator high limit switch, open on temperature rise, set to open at 42°F and set to close at 40°F.
b. Refrigerator low limit switch, open on temperature drop, set to open at 33°F and set to close at 34°F.
c. Freezer high limit switch, open on temperature rise, set to open at 0°F and set to close at -2°F.

When the refrigerator is specified for explosion-proof applications, the temperature alarm switch shall be installed in explosion-proof enclosures as specified in 3.5.3.3.2.

3.5.9 Identification plates. Each refrigerator and cabinet shall be provided with an identification plate. The size of the letters, numerals, and other characters shall be as necessary for readability. The plates may be etched, engraved, or cast using either nickel copper alloy, brass, corrosion-resisting steel, or aluminum alloy, or may be of the anodized aluminum photographic process. In addition to the identification plate, equipment for explosion-proof applications shall have a warning plate or decal. The warning plate or decal shall be located on the front of the junction box. Lettering shall be readily legible, reading substantially as follows:

CAUTION: DISCONNECT POWER SUPPLY BEFORE OPENING ELECTRICAL BOX COVERS. DO NOT ENERGIZE UNLESS THE EQUIPMENT IS FULLY ASSEMBLED AND ALL ELECTRICAL ENCLOSURES ARE PROPERLY CLOSED

3.5.9.1 Anodized aluminum photographic plates. Anodized aluminum identification plates shall conform to type H as specified in MIL-P-15024/6. The photosensitive silver compounds shall be embedded within the oxide layer. After photographic processing, the image shall be sealed within the oxide layer by anodizing in accordance with type I, class 1 as specified in MIL-A-8625. Stamping of additional marking information after anodizing shall not be permitted.

3.5.9.2 Plate nomenclature. The identification plate shall contain the following information:

1. National Stock Number (NSN).
2. Manufacturer's nomenclature and identifying number.
3. Serial number with assigned suffix letter, when required by the contract.
4. Contract number (procurement document identification number).
5. Government ownership designation (US).
6. Special characteristics, if applicable.
7. Manufacturer's code.

3.5.9.3 Plate location. The identification plate shall be located in a conspicuous place readily seen from the front of the unit or when door is opened on the right hand upper side wall.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

3.6 Workmanship. The units shall be free from fractures, splints, punctures, tears, dents, creases, bows, miscasts, deteriorations, and malfunctions. There shall be no sharp edges, burrs, or slivers. The finish shall be free from smudges, poor touch up, scale, scratches, stain, overspray, sag, imprint, or embedded foreign material. Fasteners shall not be broken, stripped, or loose.

3.6.1 Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be continuous, sound, smooth, and free from porosity, cracks, imcomplete fusion, and deformation of material. All scale and flux deposits shall be removed from the finished welds and rough edges and projections shall be dressed smooth.

3.6.2 Electrical wiring. Wiring shall not be cut, abraded, or have excessive insulation removed and shall be properly and tightly joined at terminals. Wiring shall have proper slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certification. Certification, signed by a responsible official, that no mercury containing instruments have been used in the manufacture or testing of any item furnished in accordance with this document.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 In-process inspection.

4.2.2.1 Insulation density testing. Prior to beginning each day's production of refrigerators, frozen food cabinets, or refrigerator-frozen food cabinets, one sample of the insulation to be used in that day's production shall be tested as specified in 4.4.9 for conformance to the applicable density requirement specified in 3.2.6.1 or 3.2.6.2. Any test failure shall be cause for rejection of the insulation lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

4.2.3 Representative item inspection. One representative item of the specified type and style shall be inspected for conformance to the size and capacity requirements specified in table I through VI, as applicable and tested as specified in table VII. Any nonconformance to size or capacity requirements or any test failure shall be cause for rejection of the lot. If the representative item passes all tests, subsequent lots produced using the same design, components, and manufacturing procedures, may be accepted without repetition of the test specified in table VII. Any change in design, component, or manufacturing procedure shall be cause for selection of another representative item and for inspection as specified above.

TABLE VII. Representative item performance tests

Characteristic	Requirement paragraph	Test paragraph
Condensation	3.2.6	4.4.1
Inclination	3.4.3	4.4.2
Explosion-proof	3.4.4	4.4.3
Operating requirements	3.5.8	4.4.7
Overload capability	3.5.8.3	4.4.8

4.2.4 End item visual examination. The end item shall be examined for the defects listed in table VIII. The lot size shall be expressed in units of end items of one type, style, and class only. The sample unit shall be one completed item. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for total (major and minor combined) defects.

TABLE VIII. End item visual defects

Examine	Defect	Classification Major	Minor
Seal caps	Gasket missing		X
	Not tight		X
Cold lines	Insulation missing	X	
Door latches	Do not operate properly	X	
Door gaskets	Not properly seated	X	
	Do not seal	X	
Workmanship	Part missing	X	
	Any component fractured, split, or malformed		X
	Burrs or sharp edges		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

TABLE VIII. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Welding	Missing, incomplete, cracked, fractured, or not fused	X	
	Undercut, not smooth and uniform, or scale or flux deposits not removed		X
Fasteners	Not tightened		X
	Not plated as applicable	X	
Corrosion protection	Missing, inadequate, other than specified	X	
General assembly	Component incorrectly located	X	
	Component missing	X	
Finish	Other than specified	X	
Identification plate	Missing, not as specified, not in a conspicuous location, or does not contain required information	X	
Electrical assembly	Wiring cut, abraded, or not properly installed or joined	X	
	Adequate slack not provided for wriing to relieve strain		X
Warning plate or decal (when applicable)	Missing, not legible, not located as specified, or does not contain information as specified or required	X	

4.2.5 End item testing. Each end item shall be subjected to the tests specified in 4.4.4 thru 4.4.6. Any end item unit failing one or more tests shall be rejected.

4.2.6 Packaging inspection. The sampling and inspection of the preservation and packaging shall be in accordance with the requirements of MIL-R-12323.

4.3 Certificate examination. Certificates of compliance, certified test reports, or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the document requirements, shall be examined and validated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

4.4 Methods of inspection. During the tests involving performance, the ambient temperature and the electrical voltage and current shall be recorded. Readings shall be taken at intervals of not more than 20 minutes for the duration of the test involved.

4.4.1 Condensation test. The unit shall be tested by operating for 3 hours after establishment of design operating conditions with an ambient temperature of 100°F and an ambient relative humidity of not less than 50 percent. During the test, there shall be no visible evidence of moisture forming on the external cabinet due to condensation or dripping on door mullions of multiple door cabinets.

4.4.2 Inclination test. The unit shall be inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other and operated at least 1 hour in each plane with no abnormal variations in temperature or performance.

4.4.3 Explosion-proof test. When required, explosion-proof testing shall be in accordance with UL 886 and the applicable provisions of NFPA No. 70.

4.4.4 Pressure and leak test. All refrigerant containing parts on the high pressure side of the refrigeration system shall withstand, without damage, an air and gas test of 300 pounds per square inch (psig) for R-22 and 235 psig for R-12 refrigerants. The refrigerant containing parts on the low-pressure side of the refrigeration system shall withstand without damage an air and gas test of 150 psig for R-22 and 140 psig for R-12 refrigerants. While subjected to the pressures indicated above, the refrigerating system shall be leak tested by means of a halide leak detector and shall be free of evidence of leaks. The halogen leak detector shall have a leak index sensitivity of 1/2 ounce per year.

4.4.5 System integrity test. The unit shall be operated for a period of at least 1 hour with controls set for specified operating conditions. At the conclusion of this operational test, the entire refrigerant circuit connections under refrigerant pressure shall be given a halide test to determine leakage. During this test it shall be verified that the controls are adjusted and functioning properly.

4.4.6 Door sealing test. The proper seating and sealing of the entire door gasket in refrigerators and frozen food cabinets shall be tested. A 0.003 inch thickness gage, 1/2 inch wide, shall be used to determine whether the gasket around the entire perimeter of each door seats properly against both door and cabinet when door is closed and latched.

4.4.7 Operating requirements test. The unit shall be installed in 100°F ambient temperature space with the control set to maintain an average compartment temperature with no food load. The test shall start with the compartment at design temperature. The unit shall be operated for a period of 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

hours. Simultaneous measurements shall be made of the temperature at not less than four different points within the compartment. The average temperature shall be in compliance with this document. Under these conditions of operation, the unit shall not operate in excess of 90 percent of the time.

4.4.8 Overload test. The unit shall be installed in an 120° ambient temperature space with all doors wide open. The unit shall operate without tripping the compressor motor overload device or without breakdown. The test shall be continued until steady conditions have been observed for at least 4 hours. At the end of the 4-hour test period the current shall be switched off for 10 seconds and then reapplied to permit operation. When the current is reapplied, the unit shall resume normal operation within 5 minutes. The temperature rise of the compressor motor winding shall be determined at the end of this period. The temperature rise shall be not greater than that specified in 3.5.3.1. The temperature rise of the running winding in the compressor motor shall be measured and computed by the resistance method.

4.4.9 Density test for insulation. When pre-expanded insulation is used, a 6-inch cube shall be cut and weighed. When foamed-in-place insulation is used, one free foam sample of the insulation shall be prepared in the form of a minimum size 8-inch cube, and a 6-inch cube shall be cut from the center of the sample and weighed. The weight of the 6-inch cube (1/8 of a cubic foot) multiplied be 8 shall be the density of the insulation sample. Any nonconformance to the density requirements in 3.2.6.1 and 3.2.6.2, as applicable, shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. The refrigeration equipment shall be preserved in accordance with the applicable requirements of MIL-R-12323, except that strainers and dehydrators shall not be coated with preservative compound.

5.1.2 Commercial. The refrigeration equipment shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A and B packing. The refrigeration equipment shall be packed in accordance with the applicable requirements of MIL-R-12323, preserved as specified in 5.1.

5.2.2 Commercial packing. The refrigeration equipment shall be packed in accordance with ASTM D 3951, preserved as specified in 5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-R-12323 or ASTM D 3951, as applicable.

5.4 Use of polystyrene (loose-fill) material. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for packaging applications such as cushioning, filler and dunnage is prohibited. When approved, unit packs and containers (interior and exterior) shall be marked and labeled as follows:

CAUTION

Contents cushioned, etc., with polystyrene (loose-fill) material shall not be taken aboard ship.
Remove and discard loose-fill material before shipboard storage of equipment.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material.

6. NOTES

6.1 Intended use. The refrigeration and frozen food equipment is for storage of perishable foodstuffs, liquids, and frozen food items requiring controlled conditions for Naval shipboard use.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type, style, size, and class required (see 1.2).
c. If the condenser unit for style 2 is other than left side (see 3.4).
d. Power supply required (see 3.5.3).
e. Type of compressor motor enclosure required (see 3.5.3.1).
f. Type of controller enclosure required (see 3.5.3.1.1).
g. If other than corrosion-resistant steel cabinet is required (see 3.5.4).
h. Material of refrigerated compartment (see 3.5.4).
i. Location of door hinges (see 3.5.4.3).
j. When temperature alarm actuating switches are required (see 3.5.8.5).
k. Selection of applicable levels of preservation, packing and marking (see 5.1, 5.2 and 5.3).
l. When the use of polystyrene material is approved by the procuring activity (see 5.4).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA

Review activities:

Army - MD, TS
DLA - GS

Preparing activity:

Army - GL

Project No. 4110-0377

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-21098E	Refrigerators;Frozen Food Cabinets; and Combination Refrigerator-Frozen Food Cabinets, Mechanically Refrigerated, Commercial , Self-Contained, Naval Shipboard

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 02 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 20 Nov 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:53Z
Check the source to verify that this is the current version before use.